

Measuring the External Field

A cortical-proxy content observable and the
mean-field bound on media-driven opinion
change

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Physics has always modelled media as a **field** pushing on a population.

Nobody measures the field. They assume it.

I built an instrument that measures one, ran it on 400 articles, and worked out how strong the coupling would have to be for a spread that size to move anyone.

01

The problem

Two articles, one event, two different readers.

The difference everybody notices and nobody measures

WIRE REPORT

Federal Reserve holds interest rates steady, citing a stable inflation outlook.

informs

THE SAME EVENT

FED DESTROYS AMERICA.
Your savings are GONE. The collapse they hid from you!

provokes

Sentiment analysis counts affective words. Readability counts syllables. Stance detection classifies a position.

None of them measures what the content is predicted to *do* to the person reading it.

So measure the response, not the words



A released neural encoder predicts cortical activity from the content. The measurement is taken *on that prediction*.

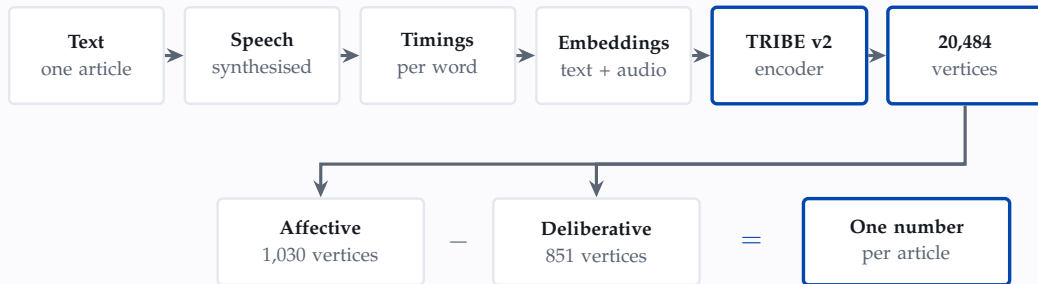
The instrument is a thermometer for one specific property of media. This project is an account of building it, running it, and saying carefully what its readings support.

02

The instrument

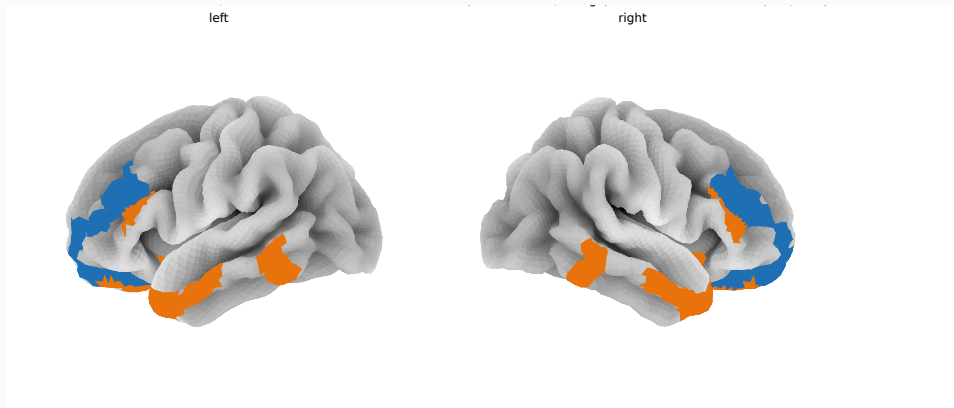
From a passage of text to a single number.

The cascade



Running publicly at `monarch-4iy.pages.dev`

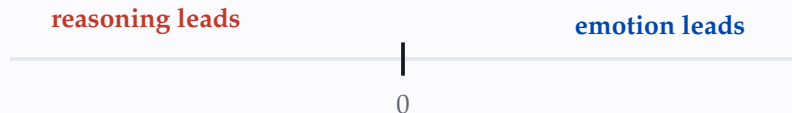
The two regions it reduces over



Affective salience in orange, deliberative control in blue. Resolved once into vertex index sets and cached, so every one of the 400 items is averaged over identical regions.

The observable

$$NAA_{\text{signed}} = A_{\text{aff}} - A_{\text{del}}$$



A **cortical proxy**. It rates content, never a person, and is never called the amygdala.

Three things the released model turned out to be

Established by inspecting the artifact, not by reading the paper about it.

01

Cortical only

Output is $(T, 20484)$. There is no subcortical head.

The amygdala is not predicted, so the proposal's central equation cannot be computed at all.

02

Averaged over subjects

The loader forces `average_subjects`.

It predicts a typical viewer. Whether that carries signal at all is the third paper's question.

03

Standardised output

Region means sit near zero, so a *ratio* sign-flips or explodes.

Undefined for 69 of 400 items. Replaced by the difference.

Each is a property of the instrument, found by testing it, and each is stated in the abstract rather than the limitations.

03

The measurement

400 articles, scanned twice.

The corpus

100 fear-activating ISOT-fake	100 high outrage SemEval-2019 Task 4	100 reward hook Webis-Clickbait-17	100 neutral in- formational PubMed + ISOT-true
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Matched on length. Mean word counts 167.4, 163.5, 163.7, 162.2, standard deviations near 11. Length is controlled and is not what drives any result.

Powered before the scan. Smallest detectable $\eta^2 = 0.0268$; smallest detectable AUC = 0.5916. Both fixed before a single item was run.

Verified row by row against the source corpus before any analysis.

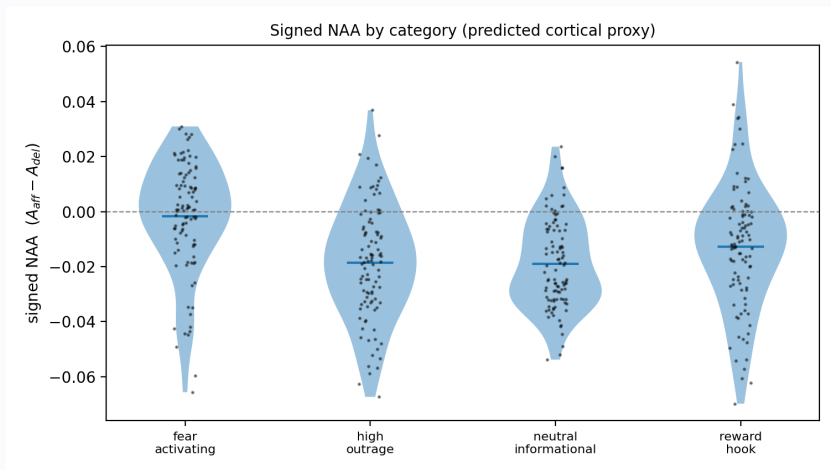
THE FOUR CATEGORIES SEPARATE AT

$$\eta^2 = 0.1068$$

$$F = 15.779, \quad p = 1.03 \times 10^{-9}, \quad n = 400$$

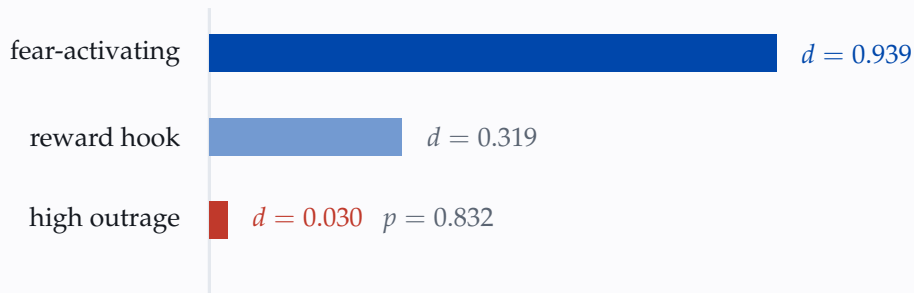
against a design powered to detect 0.0268

Every item, by category



A shift between overlapping distributions, not a clean split. Every category mean is negative: the deliberative network leads throughout.

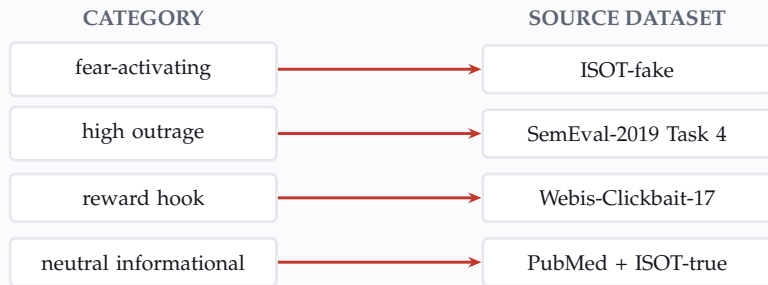
The separation is one category



Cohen's d against the neutral baseline. Fear-activating content carries almost the whole effect.

High outrage does not separate at all — the category the proposal expected to separate *most*.

The confound, said before the result



Each category is drawn from **exactly one** source. So the separation is between *corpora* and cannot be told apart from a separation between *framing styles*.

Length is controlled; provenance is not. The one piece of evidence against a pure source artifact is internal: high outrage has its own source and does not separate.

Scanned twice, so reliability is measured

0.8725

ICC between two GPU sessions

The separation replicates: $\eta^2 = 0.0888$ on the second session against 0.1068 on the first. Both clear the detectable floor.

Group-level claims hold.

The measurement

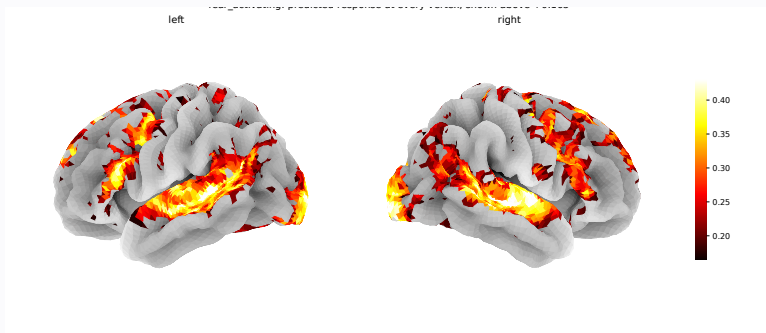
12.8%

of items reverse direction

51 of 400 items flip sign between sessions. A statement that one particular article is affective-leading is not supportable at this precision.

No per-item verdict appears anywhere — not in the analysis, the figures, or the public site.

One article, at every vertex



The encoder's own output for one fear-activating item, all 20,484 vertices, painted above the map's 70th percentile so the anatomy stays readable. Nothing is smoothed, interpolated or simulated.

04

The physics

What a measured field is worth.

Media as a field on a coupled population

Each person holds an opinion $s_i = \pm 1$, is pulled toward their neighbours with strength J , and is pushed by media with field h . Mean-field self-consistency:

$$m = \tanh(\beta J m + h), \quad h = \alpha X$$

where X is the measured observable and α is the unknown coupling.

Expanding $\operatorname{artanh}(m) \approx m + m^3/3$ gives the Landau free energy

$$F(m) = am^2 + bm^4 - hm, \quad a = \frac{1 - \beta J}{2}, \quad b = \frac{1}{12}$$

Coefficients **derived** from the self-consistency condition. The proposal carried $a = 1 - \beta J$, $b = (\beta J)^3/3$, which do not follow from it. The solver is checked against the exact exponents $\beta = 1/2$, $\gamma = 1$, $\delta = 3$.

Fitting the coupling fails, and that is informative

What was tried. Regress the outcome on the measured index and read off α .

What happened. The fit performs *worse than the sample mean* out of sample, and the estimate scales inversely with a coupling βJ that no measurement here fixes.

So no value of $\hat{\alpha}$ is quoted. Anywhere. Not in a table, not in a caption, not for illustration.

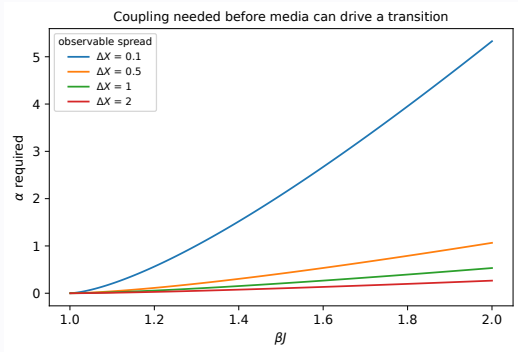
Reporting an unidentified coupling as though it were a measured one would be the failure here. Declining to quote it is not.

What replaces it is a statement that does not need the fit at all.

The bound: what any content observable must satisfy

$$\alpha \geq \frac{h_c(\beta J)}{\Delta X}$$

No transition is possible unless the coupling clears this. Rather than fitting α and hoping it clears zero, it states what α *would have to be* for the mechanism to work at all — for **any** candidate observable, before a corpus exists.



Measured spread $\Delta X = 0.1241 \Rightarrow \alpha \geq 0.169$ at $\beta J = 1.10$, 1.655 at 1.50, **4.294** at 2.00

05

Standing

What holds, and what is still open.

Does the encoder track real cortex at all?

A commercial lab reports that the released checkpoint, in the averaged configuration this project runs, is **anti-correlated** with real cortex. Four subjects, one stimulus, not independently replicated.

0.1517

measured noise ceiling

CI [0.1484, 0.1551], 4
subjects, 1000 parcels

0.2247

withdrawn

an episode picked by position,
and a guard that could not fire

Replaying the old code path reproduces the wrong number *exactly*, which is what proved the cause. Both defects were silent: nothing crashed.

The prediction pass is specified in the paper and is on its final run.

What is claimed, and what is not

CLAIMED

- A content observable that can actually be measured, with its reliability stated rather than assumed.
- That it separates these four corpora at $\eta^2 = 0.1068$, with the power statement attached.
- A bound any candidate field observable must satisfy, usable before data is collected.

NOT CLAIMED

- That the index detects manipulation. Category is confounded with source.
- Anything about the amygdala. The checkpoint does not predict it.
- Any value of the coupling α . The corpus does not identify it.
- Any verdict on one article. 12.8% reverse between sessions.

What comes out of it

72 pp

Dissertation

Complete draft.

12 pp

Paper 1

The bound and the screening criterion.

Physica A. Independent of the scan.

12 pp

Paper 2

The instrument, the corpus, the field bound.

Physica A.

6 pp

Paper 3

The measured ceiling and the replication. *Imaging Neuroscience*. One computation on its final run.

Every number in all four comes from a re-runnable script. Nothing is transcribed by hand.

Public at `github.com/brn-mwai/monarch` and `monarch-4iy.pages.dev`.

Thank you

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`monarch-4iy.pages.dev`